AGV I-PRIME driverless towing vehicle

Circuit diagram

Document number «DOC-NO» · Revision A · Language en

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Documents that belong with this circuit diagram

Document	Number
Operation manual	«DOC-NO»
Component list	«DOC-NO»
Safety controller configuration report (Flexi Soft)	«DOC-NO»
Parameter list: motor drivers, scanner, line sensor	«DOC-NO»

Standards applied to this document

Standard
IEC 61082-1:2014 (presentation)
IEC 81346-1:2022, IEC 81346-2:2019 (designations)
IEC 60617 (symbols)
EN 1175:2025 (electrical installation of industrial trucks)
EN ISO 3691-4:2023, EN ISO 13849-1:2023 (safety functions)

Revision history

Rev	Change	Date	Name
A	First issue, IEC style, draft	2026-11-01	JOY M.

Legal owner

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A	First issue, IEC style, draft	2026-11-01	JOY M.

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Checked by: JOKI I.
Approved by: ROBERTA A.

Title: AGV I-PRIME driverless towing vehicle
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Page description: Cover

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Revision: A
Page: 1
Number of pages: 24

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Page	Description	Kind	Location	Safety function	Status
1	Cover	&AB			Rev A
2	Contents	&AB			Rev A
3	Designation key	&AB			Rev A
4	Symbol legend	&AB			Rev A
10	Overview: power distribution (single line)	&EFS			Rev A
11	Overview: communication (CAN, Ethernet)	&EFS			Rev A
12	Overview: safety chain	&EFS			Rev A
20	Battery, main breaker, 48 V distribution	&EFS	+BT +EB		Rev A
21	Motor drivers: power and motors	&EFS	+EB +VH		Rev A
30	24 V control supply, part 1	&EFS	+EB		Rev A
31	24 V control supply, part 2	&EFS	+EB		Rev A
32	12 V supply	&EFS	+EB		Rev A
40	Safety: emergency stop, mode selector, reset	&EFS	+EB +FR	SF1 SF4 SF5	Rev A
41	Safety: safety laser scanner, front	&EFS	+EB +VH	SF2	Rev A
42	Safety: safe torque off and feedback (EDM)	&EFS	+EB		Rev A
43	Safety controller: supply and modules	&EFS	+EB		Rev A
44	Safety controller I/O table	table	+EB		Rev A
50	Motor drivers: control connector	&EFS	+EB		Rev A
51	CAN bus	&EFS	+EB +VH		Rev A
60	I/O module: inputs	&EFS	+EB +FR +PD		Rev A
61	I/O module: outputs	&EFS	+EB +VH		Rev A
62	I/O module I/O table	table	+EB		Rev A
75	Cable overview	&MB			Rev A
80	Parts list	&PB			Rev A

Page numbers are grouped by subject, so the gaps are intentional: 1–9 front matter, 10–19 overview, 20–29 48 V power, 30–39 control supplies, 40–49 safety, 50–59 drives, 60–69 control I/O, 70–79 cables, 80–89 lists. Grey pages are planned and not part of this issue.
Kind (IEC 61355-1): &AB list of documents, &EFS circuit diagram, &MB cable diagram, &PB parts list.

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0 1 2 3 4 5 6 7 8 9

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Reference designations (IEC 81346-1:2022, classes IEC 81346-2:2019)

Form: +LOCATION -PRODUCT :TERMINAL example +EB-KE2:I5 = terminal I5 of -KE2 in the electrical box

Wires are identified by the potential or signal name on their wire markers (1P24, HWTO1, DI03). The function aspect (=) is not used.

Numbering runs through the whole vehicle per class (-SJ1 … -SJ9). The same tag is on the device label on the vehicle.

Locations (+)

Code	Location
+EB	Electrical box (harness board): breakers, converters, drivers, controllers
+FR	Front control panel: emergency stop, push-buttons, selector
+VH	Vehicle body and chassis: motors, scanner, line sensor, horn
+BT	Battery compartment
+PD	Pendant (hand-held, wired)

Classes used in this set

Class	Meaning	Tags	Check
BG	Sensing position, presence (scanner, line sensor, RFID reader)	-BG1 … -BG4	verify
FC	Protecting against overcurrent (circuit breakers)	-FC1 … -FC4	
GB	Storing electrical energy (battery)	-GB1	
GQ	Moving gases (fans)	-GQ1 … -GQ3	verify
KE	Processing signals (controllers, I/O modules)	-KE1 … -KE4	verify
MA	Driving by electric motor	-MA1, -MA2	
PH	Presenting information visually (display)	-PH1	
PJ	Presenting information acoustically (horn)	-PJ1	
QA	Switching the main power supply (supply disconnecting device)	-QA1	
SJ	Manual control (push-buttons, E-stop, selector, switches)	-SJ1 … -SJ9	
TA	Converting electrical energy (DC/DC converters, motor drivers)	-TA1 … -TA4	verify
TF	Converting signals (USB-to-CAN interface)	-TF1	verify
WD / WG	Cables: power / control and data	-WD…, -WG…	verify

verify: code taken from a secondary source; to be checked against the purchased IEC 81346-2:2019.

Potentials

Name	Meaning	IEC 60445
1P48 / 1N48	48 V bus after -QA1, + / return. Floating: not connected to the chassis	L+ / M
2P48, 3P48	48 V feeders after -FC1 (left driver), -FC2 (right driver)	L+
1P24 / 1N24	24 V control bus after -FC3	L+ / M
1P12 / 1N12	12 V bus after -FC4. 1N12 is joined to 1N24	L+ / M
PE	Chassis / protective bonding of metal parts	PE

The DC/DC converters -TA3, -TA4 are isolated: 1N48 is not joined to 1N24 / 1N12.

The 48 V system is floating: no conductor of it is connected to the chassis.

Conductor colours (IEC 60204-1 13.2.4, used as good practice; EN 1175:2025 governs)

Circuit	Colour	Type / size
48 V power: battery, breakers, drivers, converter inputs	Black	NYAF, size as drawn
24 V / 12 V control and safety wiring	Blue	AWG22 (0.34 mm²) unless drawn otherwise
Protective bonding	Green-and-yellow	Only for PE
Supplier cables (SICK, Oriental Motor, Belden)	As supplied	Core colours per the supplier

Safety functions (EN ISO 13849-1)

ID	Function	Devices	Pages
SF1	Emergency stop	-SJ1 → -KE1/-KE2 → STO of -TA1, -TA2	40, 42
SF2	Personnel detection, front	-BG1 → -KE1/-KE2 → STO	41, 42
SF3	Personnel detection, rear	Not fitted in this revision	–
SF4	Mode selection: protective field active only in AUTO	-SJ4 → -KE2	40
SF5	Prevention of unexpected start-up (reset)	-SJ3 → -KE2	40

Cross-references

/41.3 = page 41, column 3. .3 = column 3 on the same page. /41 = page 41 (not yet drawn).

Arrow at a conductor end: the conductor continues at the reference shown.

Dash-dot box: the devices inside it are in the location written at its top-left corner.

Contacts are drawn in rest position, supplies off. Red text: open item, to be resolved before release.

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Page description: **Designation key**

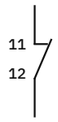
Function (=):
Location (+):
Document kind: &AB
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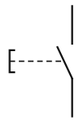
Make contact (NO)

IEC 60617 S00227. Terminal numbers per IEC 60947-5-1: 13-14, 23-24



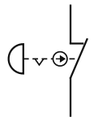
Break contact (NC)

IEC 60617 S00229. 11-12, 21-22



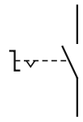
Push-button, momentary

Push operator, returns by spring



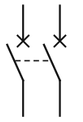
Emergency stop

Mushroom head, latching (turn to release), positive opening (IEC 60947-5-1 Annex K)



Selector switch, maintained

Turn operator, latching. Position table beside the switch: X = contact closed



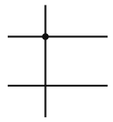
Circuit breaker

IEC 60617 S00287. Poles linked by a dashed line



Wire marker

Potential or signal name printed on the wire marker, written along the wire



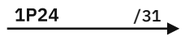
Junction / crossing

Dot = connected. Crossing without a dot = not connected



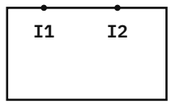
Interruption point

The conductor continues at the target and page.column shown



Potential rail

Name and source left, continuation arrow and target right



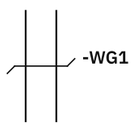
Module or device

Box with the terminals in use on its edge; tag and type inside



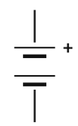
Location box (dash-dot)

Devices inside are in the location written at its top-left corner



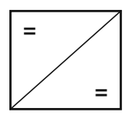
Cable

Line across the conductors of one cable, with its tag and type



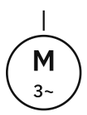
Battery

IEC 60617 S00898, + at the long plate



DC/DC converter, motor driver

Box with diagonal: input left, output right (= direct current, ~ alternating current)



Motor

Three-phase motor (brushless DC drive)



Protective earth

IEC 60417-5019. Chassis bonding point

All contacts are drawn in rest position with every supply off. Symbols per IEC 60617 (online database); JIS C 0617 is identical.

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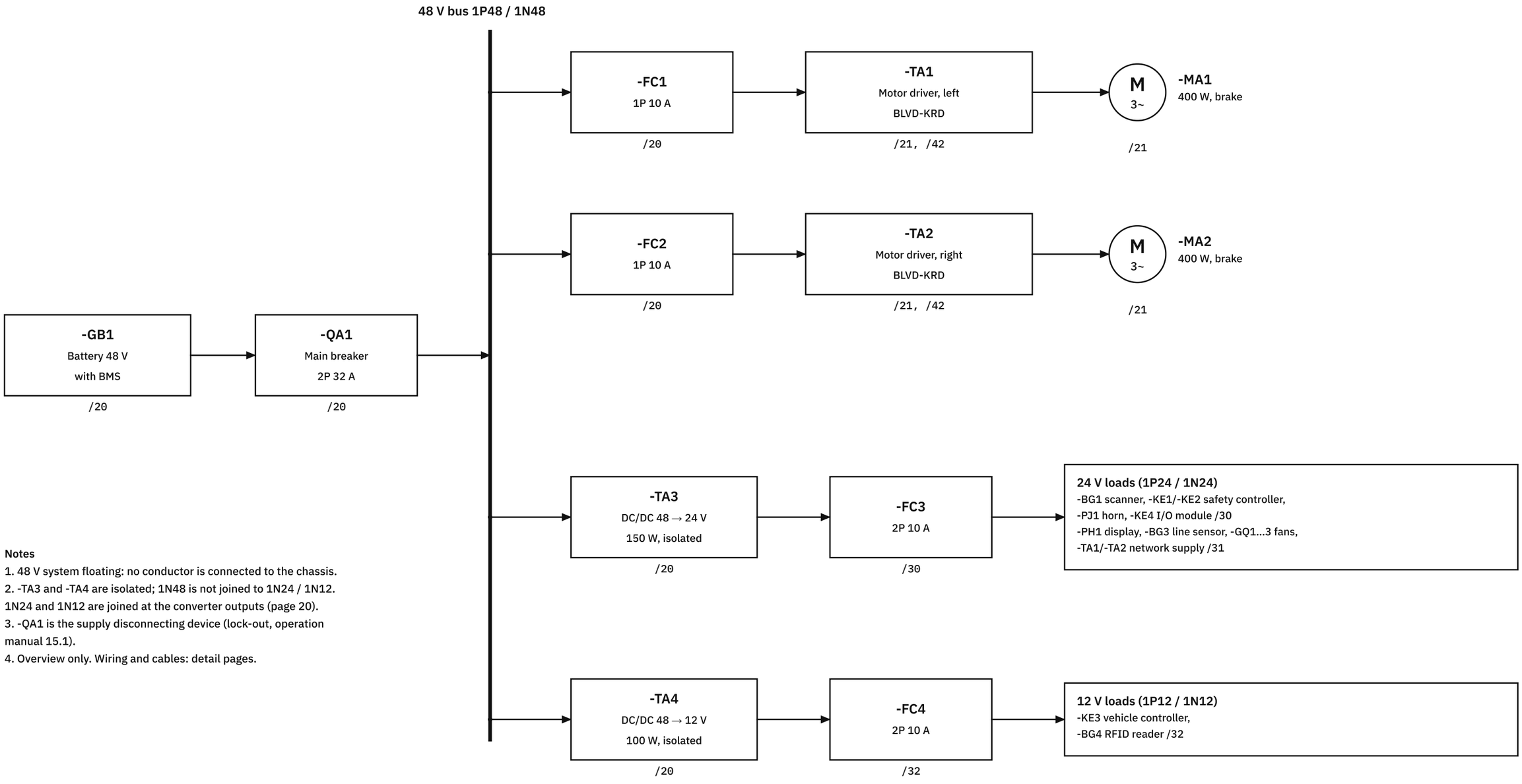
Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **Symbol legend**

Function (=)
Location (+)
Document kind: &AB
Language: en

Document number: «DOC-NO»
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Page: 4
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Notes

1. 48 V system floating: no conductor is connected to the chassis.
2. -TA3 and -TA4 are isolated; 1N48 is not joined to 1N24 / 1N12. 1N24 and 1N12 are joined at the converter outputs (page 20).
3. -QA1 is the supply disconnecting device (lock-out, operation manual 15.1).
4. Overview only. Wiring and cables: detail pages.

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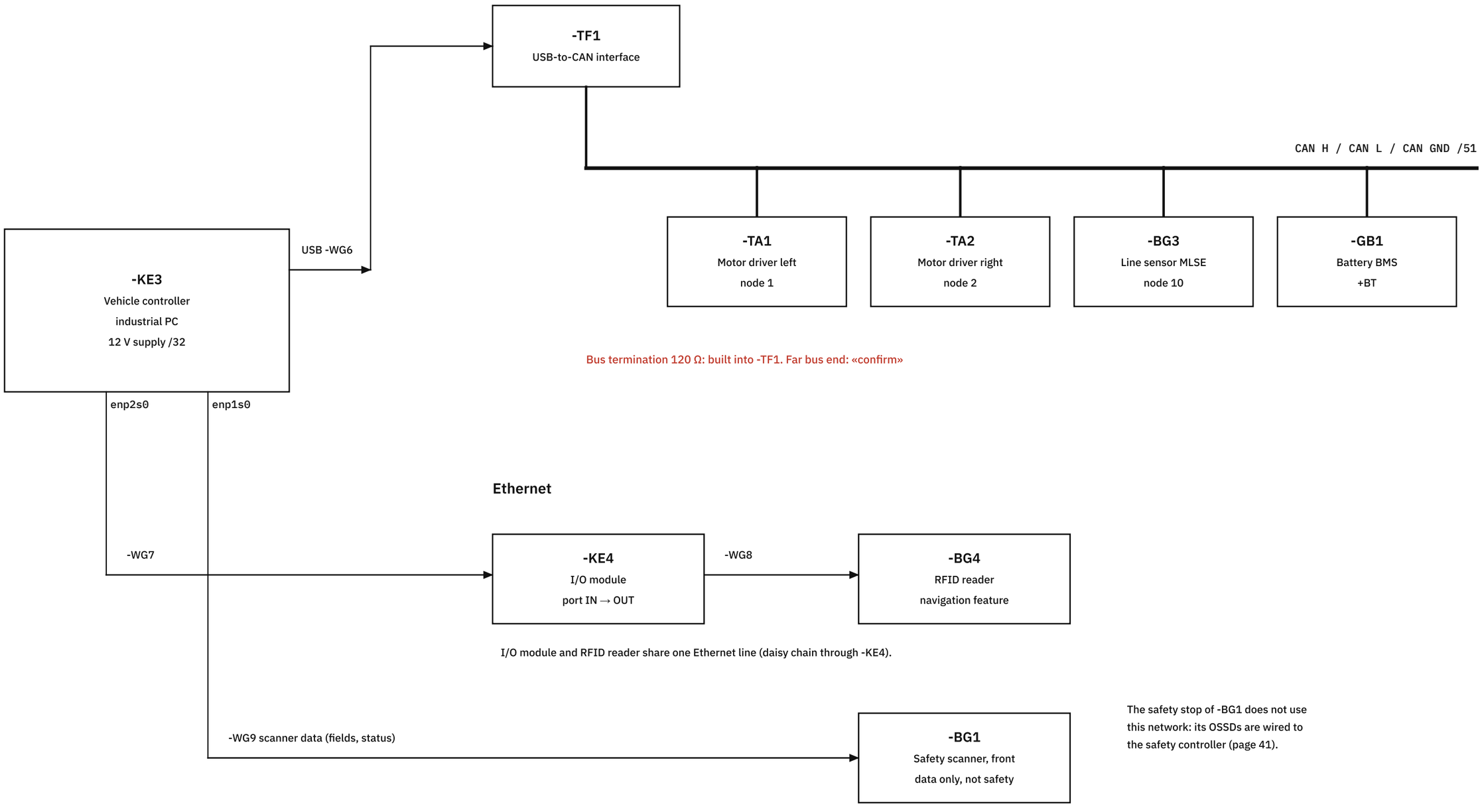
Document type: Circuit diagram (original)

Page description: **Overview: power distribution (single line)**

Function (=)
Location (+)
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Page: 10
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CAN bus, 125 kbit/s, CANopen



Bus termination 120 Ω: built into -TF1. Far bus end: «confirm»

I/O module and RFID reader share one Ethernet line (daisy chain through -KE4).

The safety stop of -BG1 does not use this network: its OSSDs are wired to the safety controller (page 41).

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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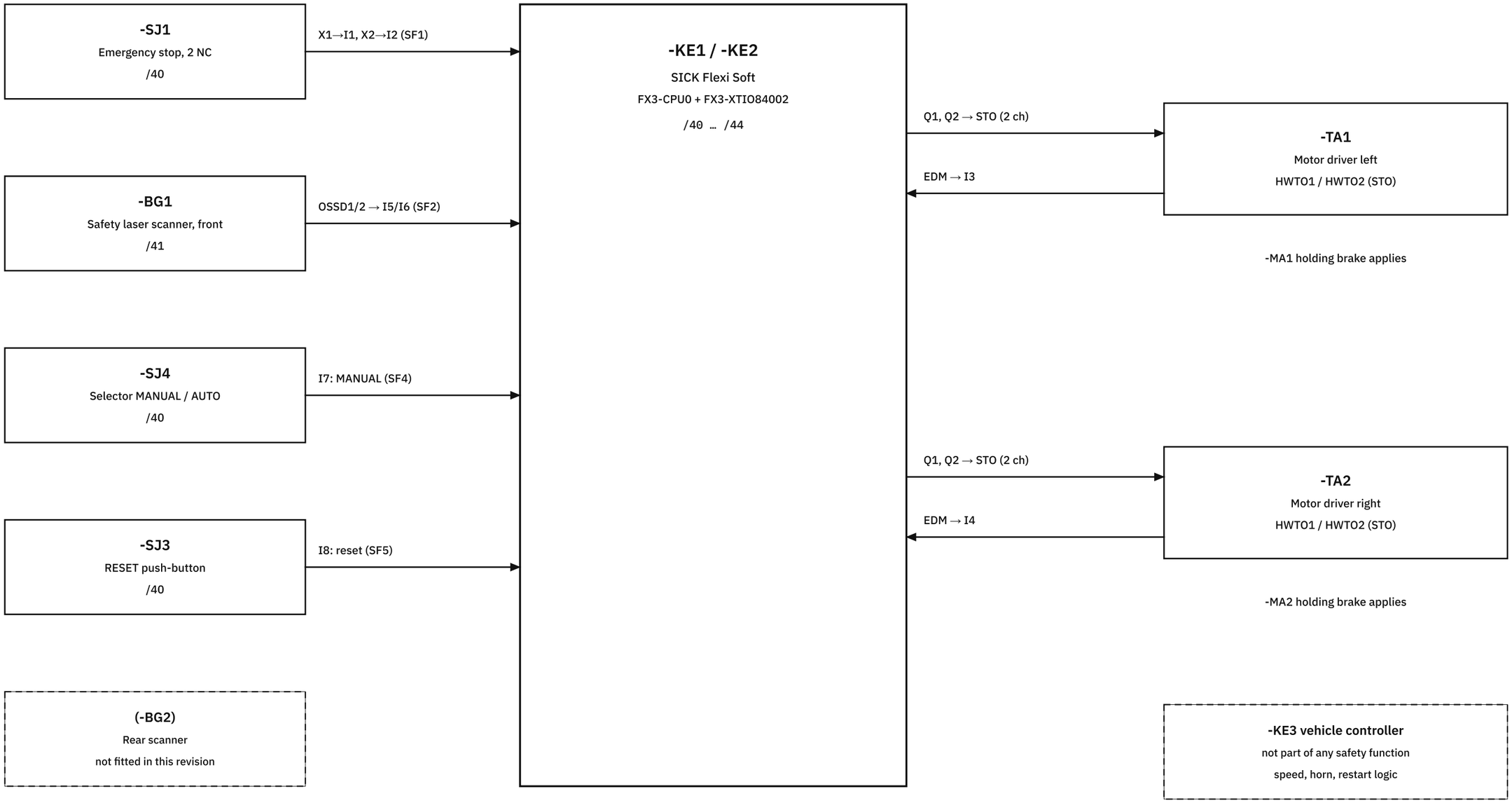
Document type: Circuit diagram (original)

Page description: **Overview: communication (CAN, Ethernet)**

Function (=)
Location (+)
Document kind: &EFS
Language: en

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SF1 emergency stop, SF2 personnel detection front, SF4 mode selection, SF5 prevention of unexpected start-up (page 3).

Stop: torque removed on both drivers (STO, driver HWTO: PL d, Cat. 3, PFHd 2.90 × 10^{-7} /h per driver), holding brakes apply.

Stop category: «0, to confirm in the risk assessment»

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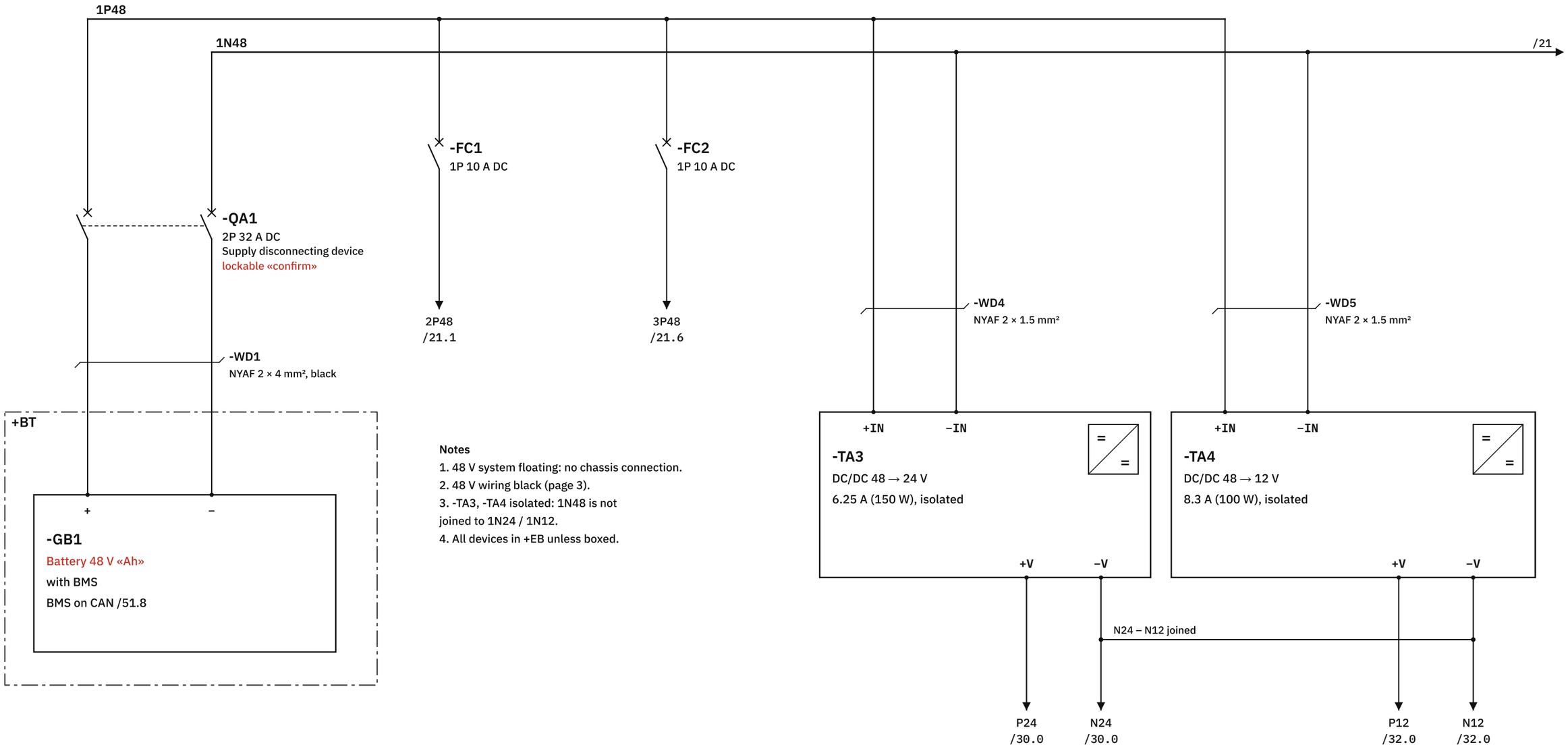
Document type: Circuit diagram (original)

Page description: **Overview: safety chain**

Function (=):
Location (+):
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **12**
Number of pages: 24

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Notes

1. 48 V system floating: no chassis connection.
2. 48 V wiring black (page 3).
3. -TA3, -TA4 isolated: 1N48 is not joined to 1N24 / 1N12.
4. All devices in +EB unless boxed.

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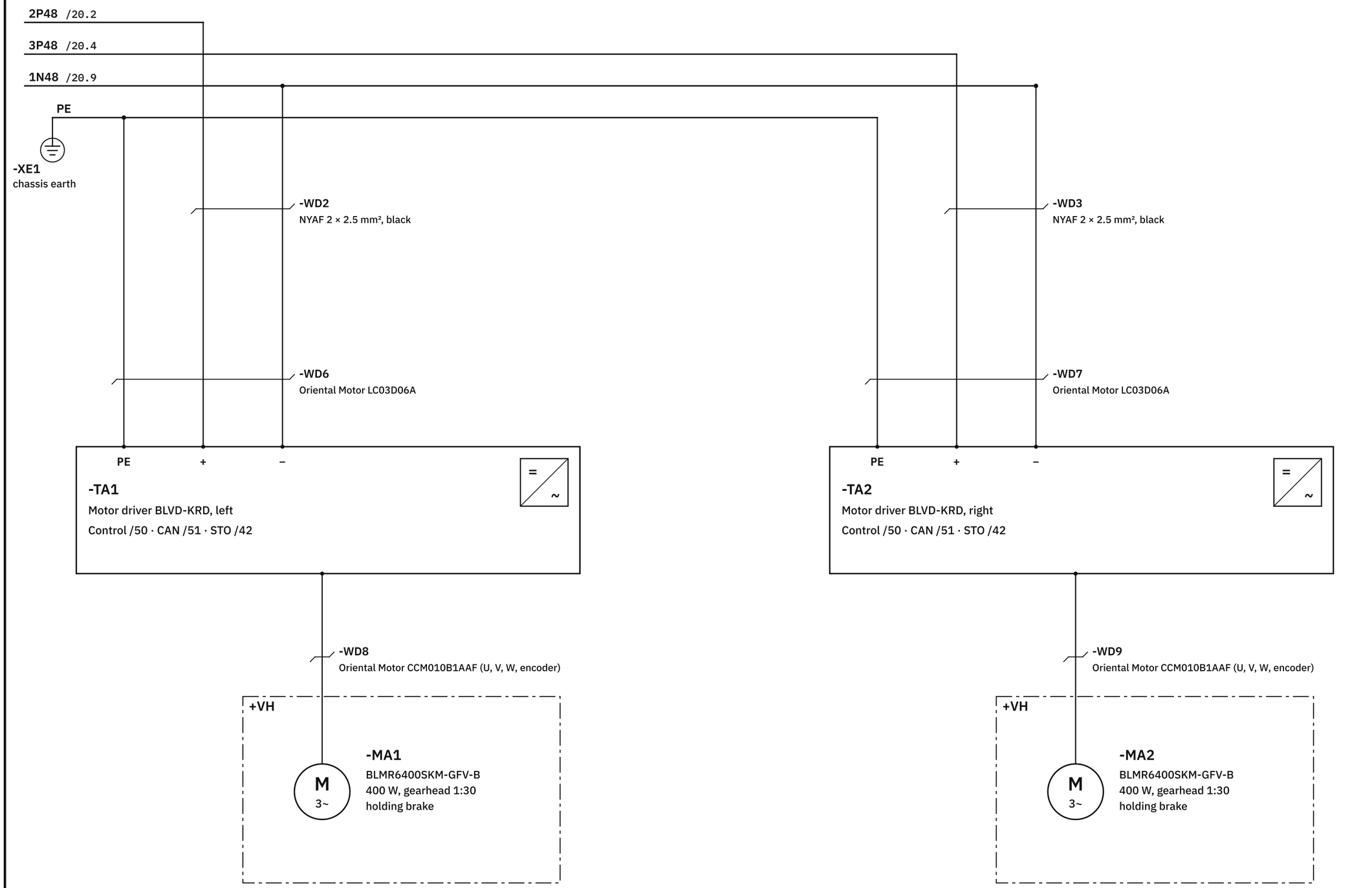
Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **Battery, main breaker, 48 V distribution**

Function (=)	
Location (+)	+BT +EB
Document kind	&EFS
Language	en
Document number	«DOC-NO»
Revision	A
Page	20
Number of pages	24

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Notes

1. 48 V wiring black.
2. PE: bonding of the driver housings to the chassis.
3. Motor and brake are fed by the driver. All their power comes in on -WD6 / -WD7 (LC03, 48 V).

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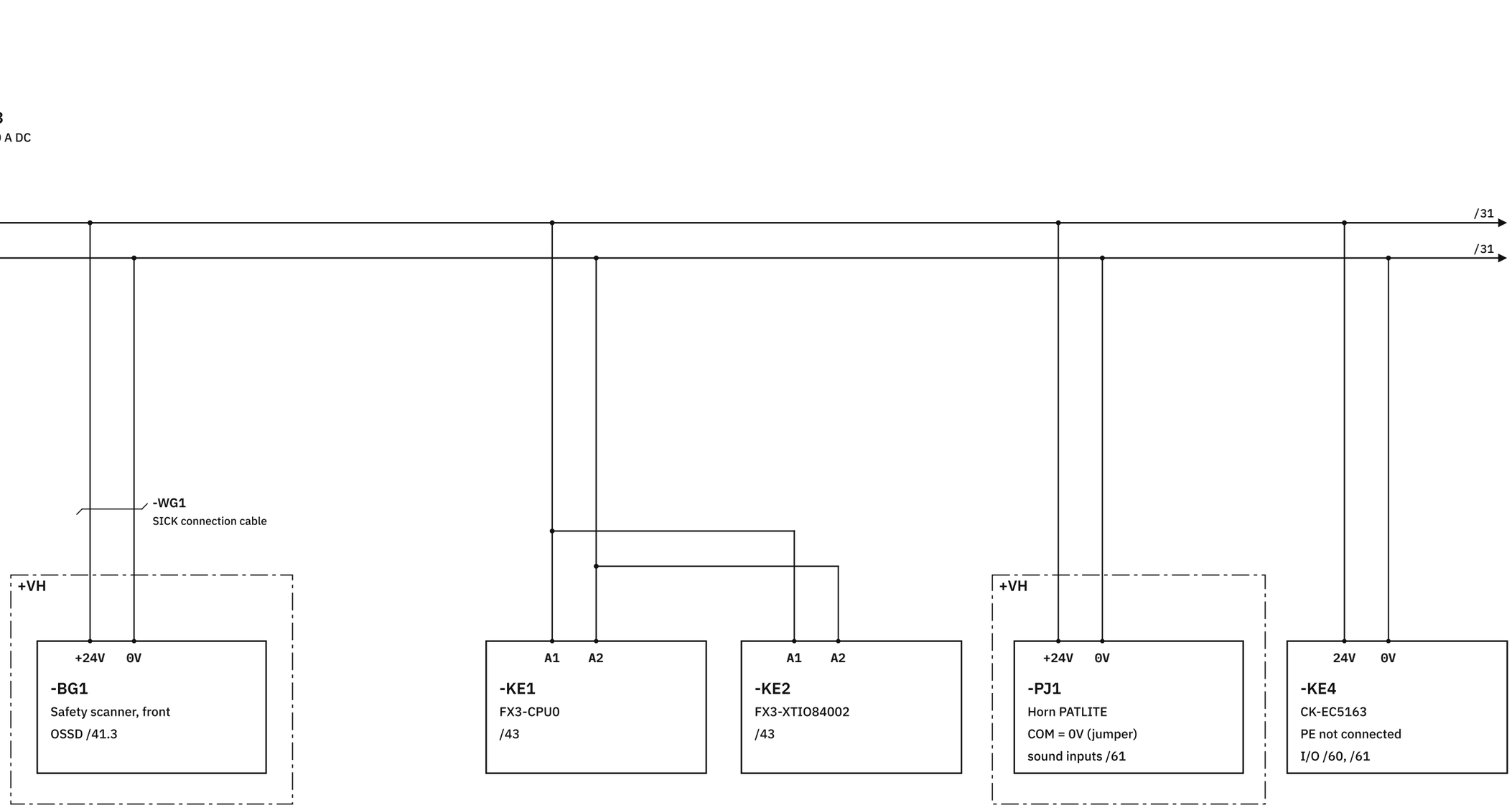
Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **Motor drivers: power and motors**

Function (=)	
Location (+)	+EB +VH
Document kind	&EFS
Language	en
Document number	«DOC-NO»
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Number of pages	24

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Notes

1. Control wiring AWG22 (0.34 mm²) blue unless marked (page 3).
2. Rear scanner (SF3) not fitted in this revision.
3. All devices in +EB unless boxed.

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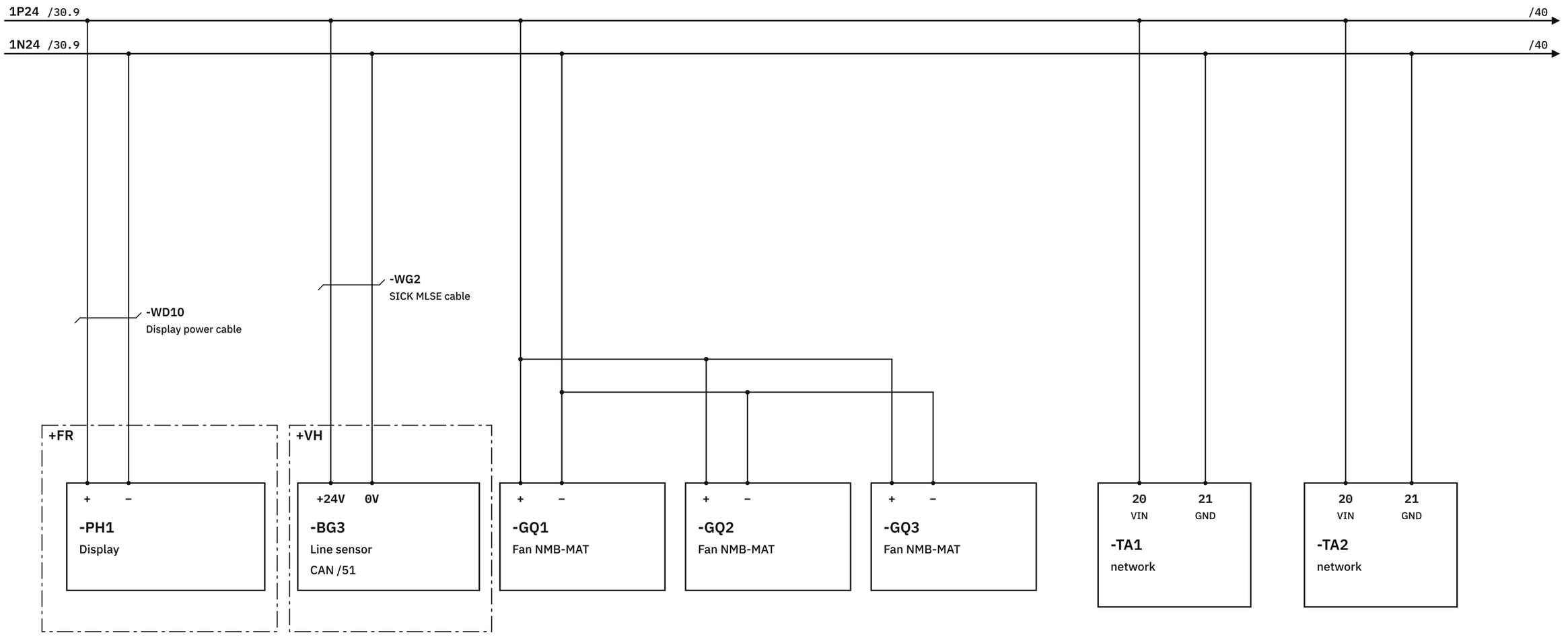
Document type: Circuit diagram (original)

Page description: **24 V control supply, part 1**

Function (=):
Location (+): +EB
Document kind: &EFS
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Document number: «DOC-NO»
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Page: **30**
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Notes

1. Control wiring AWG22 (0.34 mm²) blue unless marked (page 3).
2. -BG3 line sensor: the same cable carries CAN (page 51).
3. -TA1, -TA2: network supply only. Driver power: page 21.
4. All devices in +EB unless boxed.

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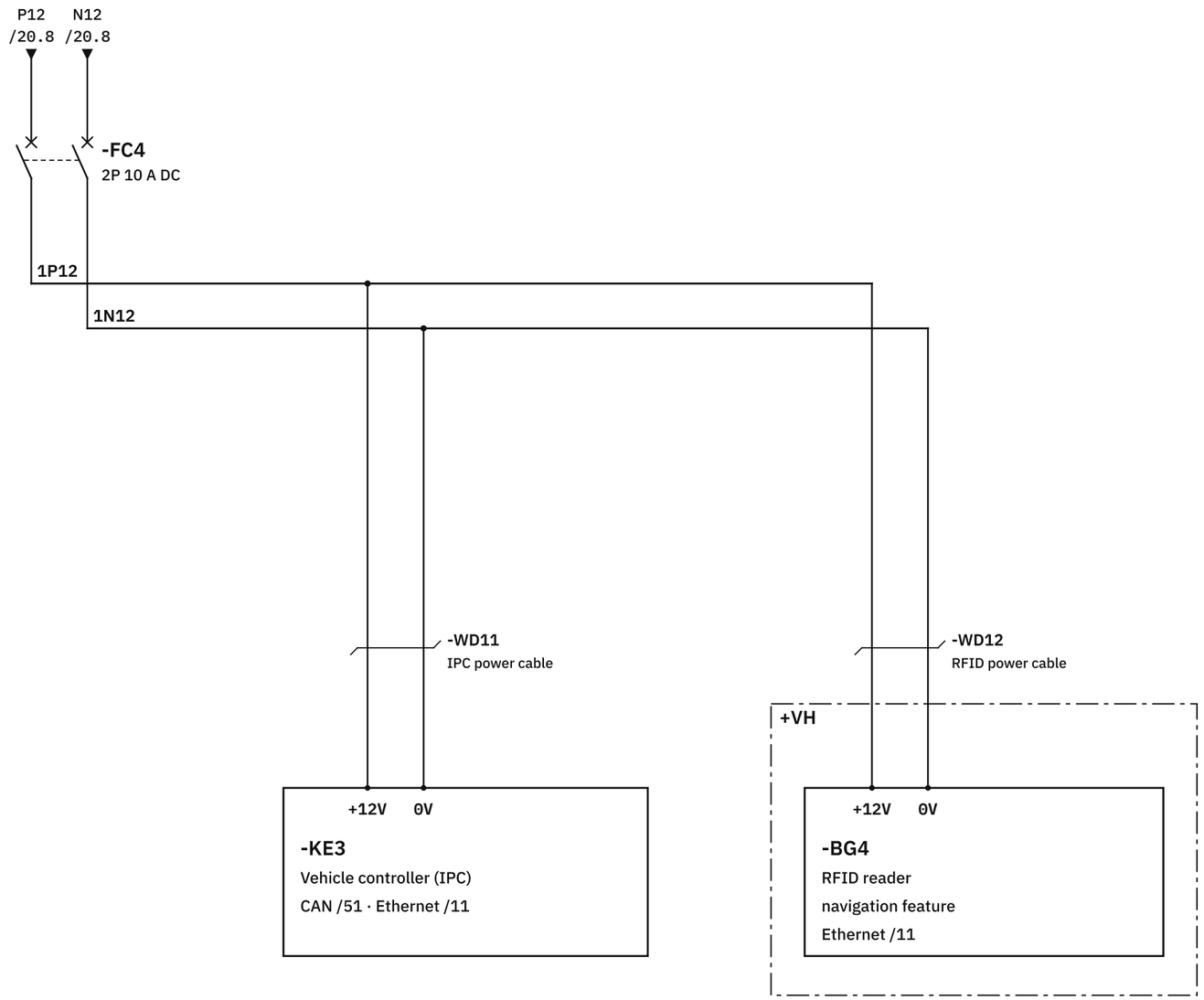
Document type: Circuit diagram (original)

Page description: **24 V control supply, part 2**

Function (=):
Location (+): +EB
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **31**
Number of pages: 24

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Notes

1. 1N12 is joined to 1N24 at the converter outputs (page 20).
2. Control wiring AWG22 (0.34 mm²) blue (page 3).
3. All devices in +EB unless boxed.

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Rev	Change	Date	Name
A	First issue, IEC style, draft	2026-11-01	JOY M.

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **12 V supply**

Function (=):
Location (+): +EB
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **32**
Number of pages: 24

1P24 /31.9 → /42

1N24 /31.9 → /42

Notes

1. Contacts drawn in rest position, supplies off. -SJ4 drawn in AUTO (contacts open).
2. X1, X2 are pulsed test outputs of -KE2. X1 → I1 and X2 → I2: odd/even pairing, SICK 8012478.
3. Wiring AWG22 (0.34 mm²) blue unless stated (page 3).

+FR front panel

-SJ1

Emergency stop
2 NC
positive opening
turn to release
SF1
«part no.»

11 12 21 22

+FR front panel

-SJ4

Selector MANUAL / AUTO
2 NO, maintained
SF4

Pos.	13-14	23-24
MANUAL	X	X
AUTO		

13 14 23 24

+FR front panel

-SJ3

RESET push-button, blue
2 NO, momentary
SF5
«part no.»

13 14 23 24

-KE4:DI02 /60.2

-KE4:DI01 /60.1

+EB electrical box

X1 I1 X2 I2 I7 I8

-KE2

SICK Flexi Soft FX3-XTIO84002 · safety controller I/O module (main module -KE1)

Other terminals: A1, A2 /43 · I3, I4 /42 · I5, I6 /41 · Q1, Q2 /42 · Q3, Q4 spare

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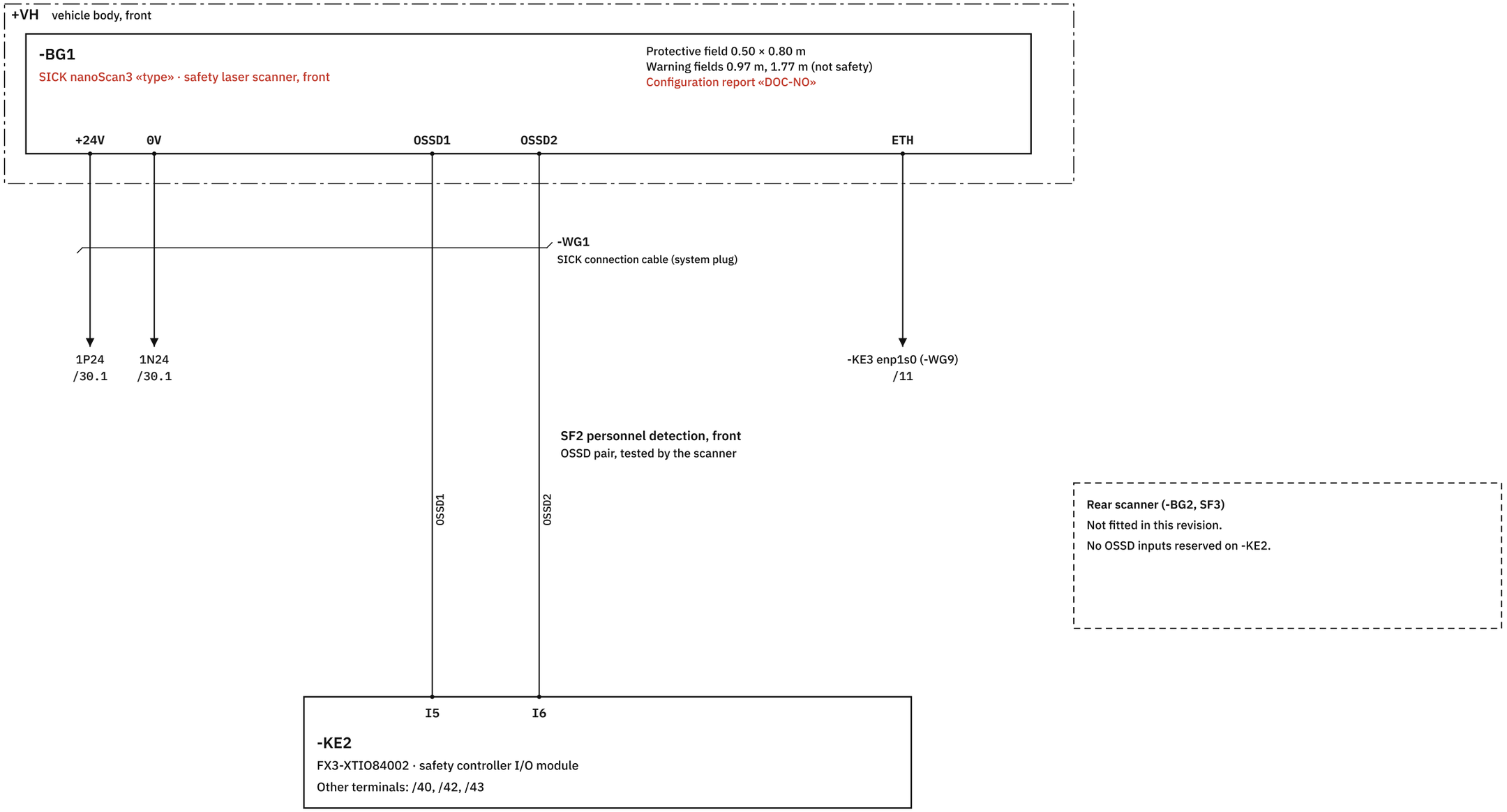
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Created by: JOY M.
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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram (original)
Page description: **Safety: emergency stop, mode selector, reset**

Function (=): SF1 SF4 SF5
Location (+): +EB +FR
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **40**
Number of pages: 24



Notes

1. -KE2 inputs I5 / I6 are configured as one OSSD input pair «configuration report».
2. The Ethernet link carries field and status data for the vehicle controller only (page 11).

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **Safety: safety laser scanner, front**

Function (=)	SF2
Location (+)	+EB +VH
Document kind	&EFS
Language	en
Document number	«DOC-NO»
Revision	A
Page	41
Number of pages	24

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0	1	2	3	4	5	6	7	8	9

1P24 /40 → /60

1N24 /40 → /60

-KE2
FX3-XTIO84002

Q1 · Q2 · I3 · I4

HWTO1 · HWTO2 · EDM-L · EDM-R

Notes

1. SF1, SF2: Q1 and Q2 switch off → torque removed on both drivers (STO).
2. Driver HWTO: PL d, Cat. 3, PFHd 2.90 × 10^{-7} /h per driver (maker data).
3. EDM: -KE2 checks that torque is really off (I3 left, I4 right).
4. HWTO1 and HWTO2 are looped from -TA1 to -TA2 (junctions).
5. Q3, Q4 spare. Test pulses on Q1, Q2: «configuration report».
6. Wiring AWG22 (0.34 mm²) blue. All devices in +EB.

11	26	14	12	27	28
HWTO1+	HWTO2+	EDM−	HWTO1−	HWTO2−	EDM+

-TA1
Motor driver BLVD-KRD, left · control connector (page 50)
Factory jumpers 11–25, 12–26, 13–27 removed

11	26	14	12	27	28
HWTO1+	HWTO2+	EDM−	HWTO1−	HWTO2−	EDM+

-TA2
Motor driver BLVD-KRD, right · control connector (page 50)
Factory jumpers 11–25, 12–26, 13–27 removed

0	1	2	3	4	5	6	7	8	9

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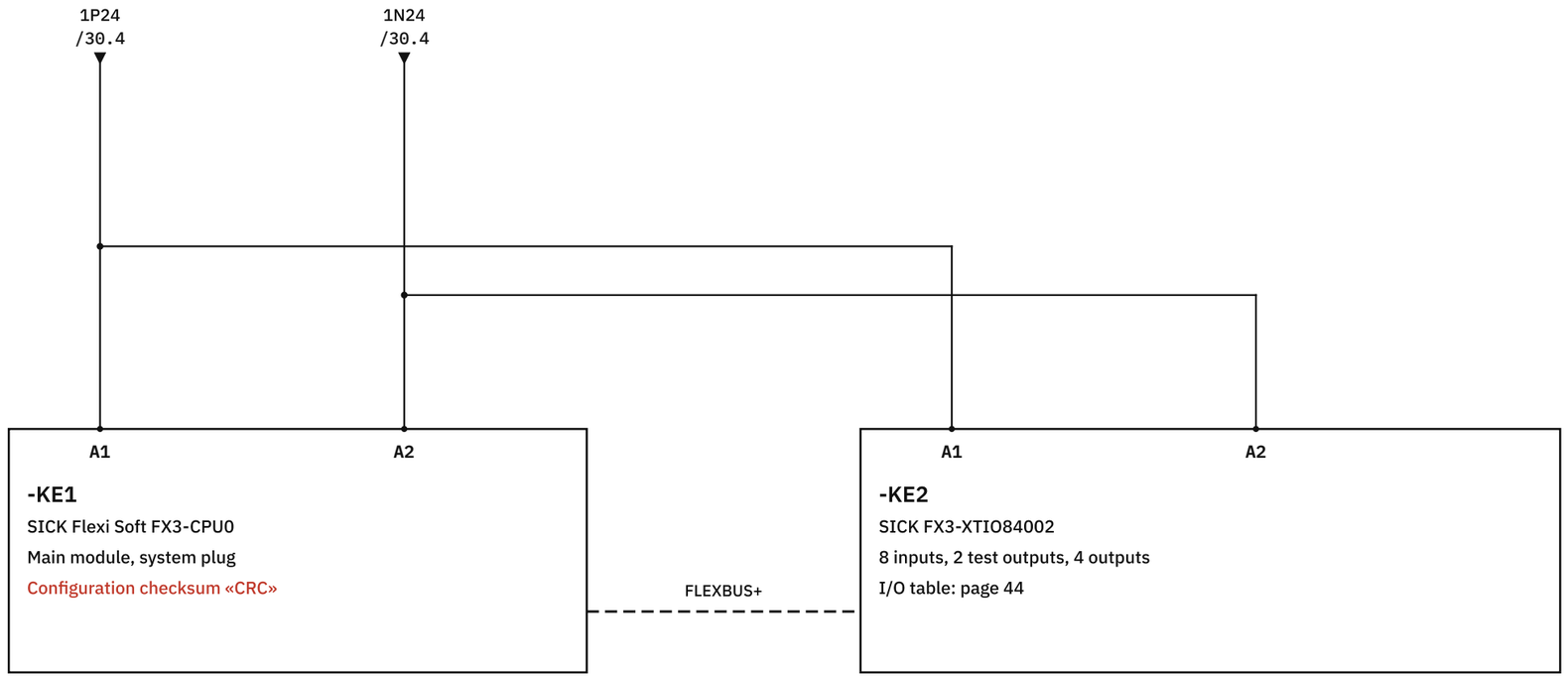
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Approved by: ROBERTA A.

Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram (original)
Page description: **Safety: safe torque off and feedback (EDM)**

Function (=):
Location (+): +EB
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: 42
Number of pages: 24

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Notes

1. Configuration: Flexi Soft Designer project «file», report «DOC-NO».
2. After any change of configuration or wiring: verify all safety functions (operation manual 8.5).
3. All devices in +EB.

-KE2 terminals: inputs

Terminal	Connected to	Page
X1	-SJ1 (11-12) → I1	40
X2	-SJ1 (21-22) → I2	40
I1	-SJ1 (11-12)	40
I2	-SJ1 (21-22)	40
I3	-TA1:14 EDM–	42
I4	-TA2:14 EDM–	42
I5	-BG1 front scanner	41
I6	-BG1 front scanner	41
I7	-SJ4 (13-14), closed in MANUAL	40
I8	-SJ3 (13-14)	40

-KE2 terminals: supply and outputs

Terminal	Connected to	Page
A1	supply	43
A2	supply	43
Q1	-TA1:11, -TA2:11 HWTO1+	42
Q2	-TA1:26, -TA2:26 HWTO2+	42
Q3	spare	
Q4	spare	

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **Safety controller: supply and modules**

Function (=):
Location (+): +EB
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **43**
Number of pages: 24

-KE2 SICK FX3-XTIO84002 — safety controller I/O module (main module -KE1 FX3-CPU0)

Terminal	Signal	Device / connection	Test pulse	Safety function	Page
A1	1P24	supply	—	—	43
A2	1N24	supply	—	—	43
X1	test output 1	-SJ1 (11-12) → I1	source	SF1	40
X2	test output 2	-SJ1 (21-22) → I2	source	SF1	40
I1	E-stop channel 1	-SJ1 (11-12)	X1	SF1	40
I2	E-stop channel 2	-SJ1 (21-22)	X2	SF1	40
I3	EDM left	-TA1:14 EDM–	«config report»	SF1, SF2	42
I4	EDM right	-TA2:14 EDM–	«config report»	SF1, SF2	42
I5	OSSD 1	-BG1 front scanner	scanner-internal	SF2	41
I6	OSSD 2	-BG1 front scanner	scanner-internal	SF2	41
I7	MANUAL	-SJ4 (13-14), closed in MANUAL	«config report»	SF4	40
I8	RESET	-SJ3 (13-14)	«config report»	SF5	40
Q1	STO channel 1	-TA1:11, -TA2:11 HWTO1+	«config report»	SF1, SF2	42
Q2	STO channel 2	-TA1:26, -TA2:26 HWTO2+	«config report»	SF1, SF2	42
Q3		spare			
Q4		spare			

Safety functions

ID	Function	Devices
SF1	Emergency stop	-SJ1 → -KE1/-KE2 → STO of -TA1, -TA2
SF2	Personnel detection, front	-BG1 → -KE1/-KE2 → STO
SF3	Personnel detection, rear	Not fitted in this revision
SF4	Mode selection: protective field active only in AUTO	-SJ4 → -KE2
SF5	Prevention of unexpected start-up (reset)	-SJ3 → -KE2

Notes

Test-pulse settings come from the Flexi Soft configuration report. They are not shown on the hardware and must be filled from the report.

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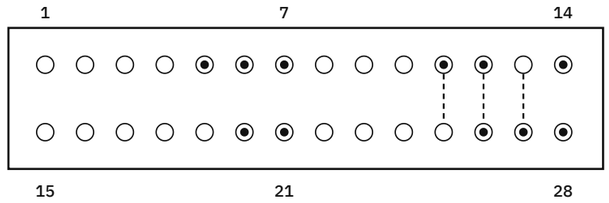
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Approved by: ROBERTA A.

Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram (original)
Page description: **Safety controller I/O table**

Function (=):
Location (+): +EB
Document kind: table
Language: en

Document number: «DOC-NO»
Revision: A
Page: 44
Number of pages: 24

BLVD-KRD control connector, 28 pins (same on -TA1 and -TA2), view on the driver



● = wired pin

--- = factory jumper, removed

Pin	Signal	Connected to	Wire marker	Page
1	OUT0+	—		
2	OUT0−	—		
3	OUT1+	—		
4	OUT1−	—		
5	CAN_L	CAN L	CAN L	51
6	CAN_H	CAN H	CAN H	51
7	CAN_GND	CAN GND	CAN GND	51
8	485GND	—		
9	TR+	—		
10	TR−	—		
11	HWTO1+	-KE2:Q1 (STO channel 1)	HWTO1	42
12	HWTO1−	1N24	1N24	42
13	0V	—		
14	EDM−	-KE2:I3 (left) / I4 (right)	EDM-L / EDM-R	42

Pin	Signal	Connected to	Wire marker	Page
15	IN-COM	—		
16	IN0	—		
17	IN1	—		
18	IN2	—		
19	IN3	—		
20	NET-VIN	1P24	1P24	31
21	NET-GND	1N24	1N24	31
22	485GND	—		
23	TR+	—		
24	TR−	—		
25	+V	—		
26	HWTO2+	-KE2:Q2 (STO channel 2)	HWTO2	42
27	HWTO2−	1N24	1N24	42
28	EDM+	1P24	1P24	42

Notes

1. Factory jumpers 11–25, 12–26 and 13–27 are removed on both drivers: the hardwired torque-off (HWTO) inputs are driven by the safety controller (page 42).
2. Grey pins are not connected. The drivers take no commands on their digital inputs; motion is commanded over CAN only (page 51).
3. Driver power: Oriental Motor cable LC03D06A (page 21). Motor and encoder: Oriental Motor cable CCM010B1AAF (page 21).
4. Driver parameters (CANopen node ID, ramps, limits): separate parameter list «DOC-NO».

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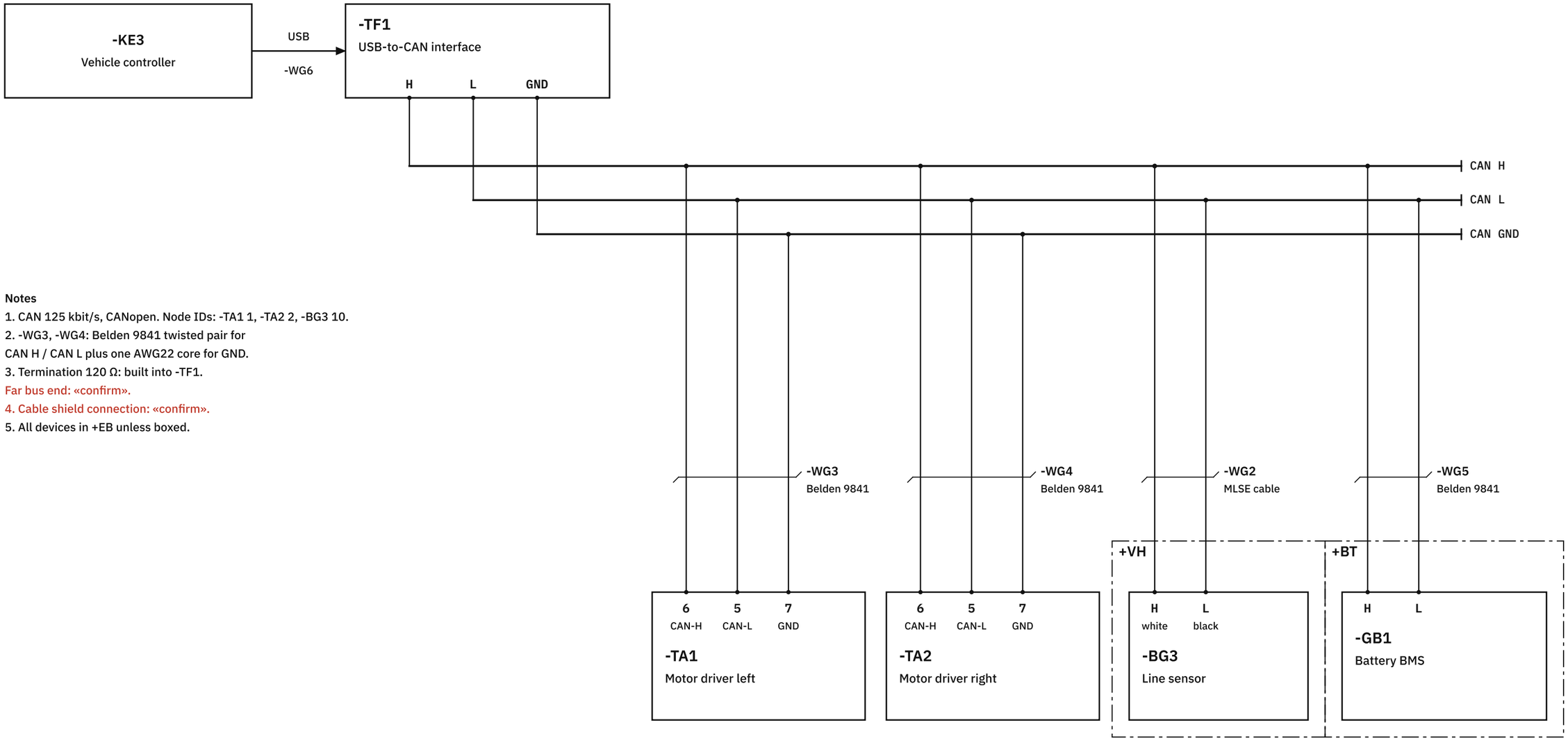
Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **Motor drivers: control connector**

Function (=):
Location (+): +EB
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: 50
Number of pages: 24



Notes

1. CAN 125 kbit/s, CANopen. Node IDs: -TA1 1, -TA2 2, -BG3 10.
2. -WG3, -WG4: Belden 9841 twisted pair for CAN H / CAN L plus one AWG22 core for GND.
3. Termination 120 Ω: built into -TF1.
   Far bus end: «confirm».
4. Cable shield connection: «confirm».
5. All devices in +EB unless boxed.

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **CAN bus**

Function (=):
Location (+): +EB +VH
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **51**
Number of pages: 24

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1P24 /42 → /61

1N24 /42 → /61

Notes

1. -SJ4 contact closed = MANUAL (DI02 on).
2. Pendant buttons: hold to run (software).
3. Wiring AWG22 blue. All in +EB unless boxed.

-SJ3:24 /40.8

-SJ4:24 /40.5

+FR

-SJ2
START
1 NO
13
14

+PD pendant

-SJ6
FORWARD
13
14

-SJ7
REVERSE
13
14

-SJ8
LEFT
13
14

-SJ9
RIGHT
13
14

-WG10
pendant cable

+FR

-SJ5
MANUAL ARM
«switch type»
13
14

DI00 DI01 DI02 DI03 DI04 DI05 DI06 DI08 COM0 COM1

-KE4

CK-EC5163 I/O module, inputs · outputs /61 · I/O table /62

DI00–DI07: common COM0 = 1P24, input on when switched to 1N24 · DI08–DI0F: common COM1 = 1N24, input on when switched to 1P24

0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **I/O module: inputs**

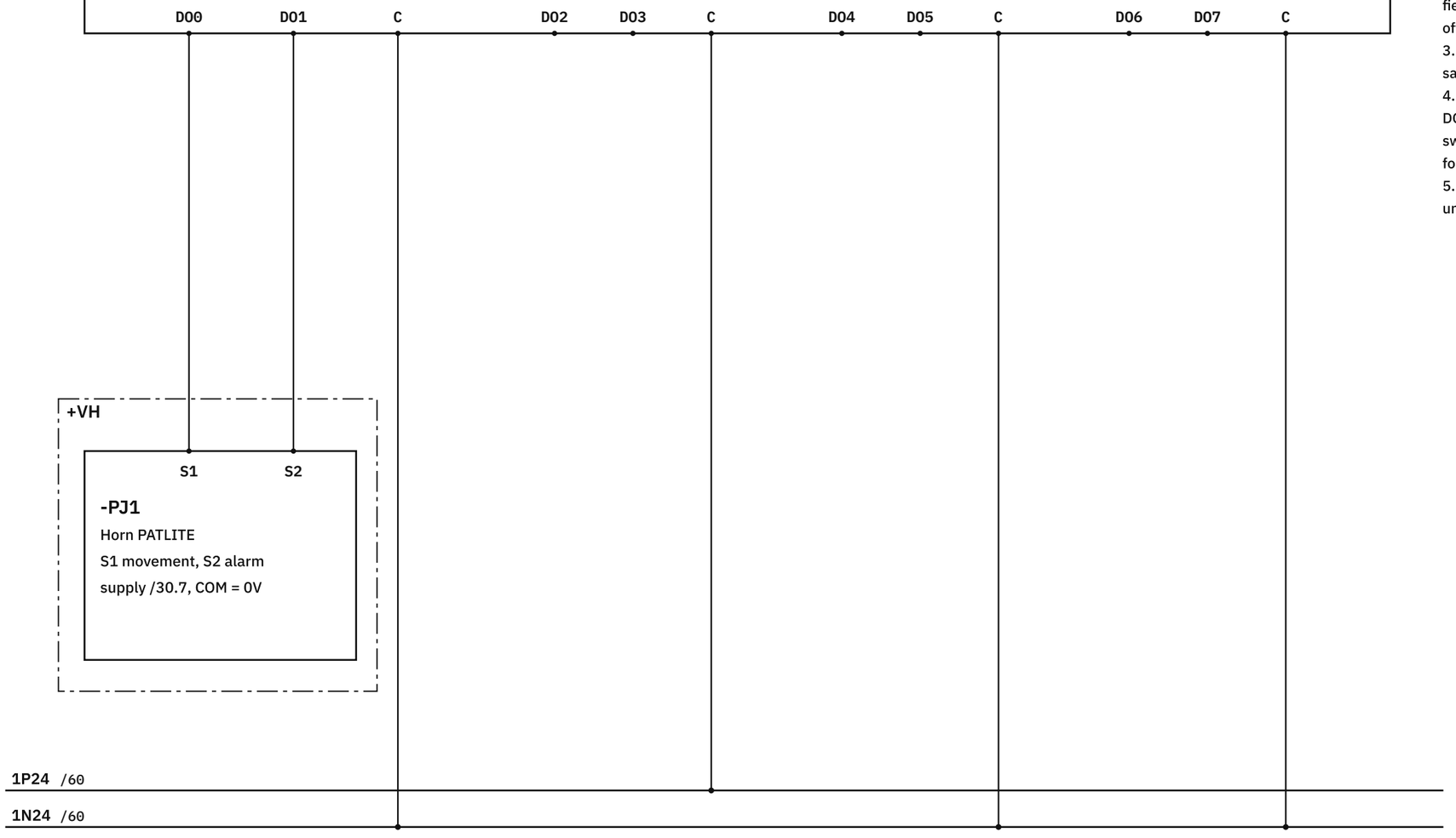
Function (=):
Location (+): +EB +FR +PD
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **60**
Number of pages: 24

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-KE4

CK-EC5163 I/O module, relay outputs: each output closes to the common C of its pair



Notes

1. DO0, sound 1 (movement): on while the wheels turn, in every mode.
2. DO1, sound 2 (alarm): 2 s after START before moving, and while a scanner field is occupied in AUTO. Sound 1 is off while sound 2 plays.
3. The horn is software-driven, not a safety function.
4. DO2 … DO7 spare; commons wired. DO8 … DOB not wired. The software also switches DO8 with motion: free output for a warning lamp.
5. Wiring AWG22 blue. All in +EB unless boxed.

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0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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Title: **AGV I-PRIME driverless towing vehicle**

Document type: Circuit diagram (original)

Page description: **I/O module: outputs**

Function (=):
Location (+): +EB +VH
Document kind: &EFS
Language: en

Document number: «DOC-NO»
Revision: A
Page: **61**
Number of pages: 24

-KE4 CK-EC5163 — I/O module, 16 DI / 12 relay DO, Ethernet (+EB)

Channel	Signal	Device	Function	Page
DI00	START	-SJ2	Start / continue automatic operation	60
DI01	RESET	-SJ3 (23-24)	Reset, second contact	60
DI02	MANUAL	-SJ4 (23-24)	ON = MANUAL, OFF = AUTO	60
DI03	P.FWD	-SJ6	Pendant forward (hold to run)	60
DI04	P.RVS	-SJ7	Pendant reverse (hold to run)	60
DI05	P.LFT	-SJ8	Pendant left (hold to run)	60
DI06	P.RGT	-SJ9	Pendant right (hold to run)	60
DI07			spare	
DI08	MANUAL ARM	-SJ5	Drives on for manual driving	60
DI09–DI0F			spare	
DO0	HORN 1	-PJ1	Movement sound: on while the wheels turn	61
DO1	HORN 2	-PJ1	Alarm sound: 2 s after START, field occupied in AUTO	61
DO2–DO7			spare (commons wired)	61
DO8			switched with motion by the software; nothing connected	61
DO9–DOB			spare, not wired	

Input commons

DI00 … DI07: common COM0 = 1P24. An input is active when it is switched to 1N24.

DI08 … DI0F: common COM1 = 1N24. An input is active when it is switched to 1P24.

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Title: **AGV I-PRIME driverless towing vehicle**
Document type: Circuit diagram (original)
Page description: **I/O module I/O table**

Function (=):
Location (+): +EB
Document kind: table
Language: en

Document number: «DOC-NO»
Revision: A
Page: 62
Number of pages: 24

Cables: power (-WD) and control / data (-WG)

Tag	Type	Cores × size	From	To	Pages	Length
-WD1	NYAF single cores, black	2 × 4 mm²	-GB1	-QA1	20	«m»
-WD2	NYAF single cores, black	2 × 2.5 mm²	-FC1, 1N48 bus	-WD6	21	«m»
-WD3	NYAF single cores, black	2 × 2.5 mm²	-FC2, 1N48 bus	-WD7	21	«m»
-WD4	NYAF single cores, black	2 × 1.5 mm²	1P48, 1N48	-TA3 in	20	«m»
-WD5	NYAF single cores, black	2 × 1.5 mm²	1P48, 1N48	-TA4 in	20	«m»
-WD6	Oriental Motor LC03D06A	3 cores (+, 0V, PE)	-WD2, PE	-TA1 power	21	«m»
-WD7	Oriental Motor LC03D06A	3 cores (+, 0V, PE)	-WD3, PE	-TA2 power	21	«m»
-WD8	Oriental Motor CCM010B1AAF	motor + encoder	-TA1	-MA1	21	1 m
-WD9	Oriental Motor CCM010B1AAF	motor + encoder	-TA2	-MA2	21	1 m
-WD10	Display power cable	2 cores	1P24, 1N24	-PH1	31	«m»
-WD11	IPC power cable	2 cores	1P12, 1N12	-KE3	32	«m»
-WD12	RFID power cable	2 cores	1P12, 1N12	-BG4	32	«m»
-WG1	SICK connection cable (nanoScan3)	«cores»	1P24, 1N24; -KE2:I5, I6	-BG1	30, 41	«m»
-WG2	SICK MLSE power & comm cable	«cores»	1P24, 1N24; CAN bus	-BG3	31, 51	«m»
-WG3	Belden 9841 (CAN H, L) + AWG22 (GND)	1 × 2 × 0.32 mm² + 1	CAN bus	-TA1:5, 6, 7	51	«m»
-WG4	Belden 9841 (CAN H, L) + AWG22 (GND)	1 × 2 × 0.32 mm² + 1	CAN bus	-TA2:5, 6, 7	51	«m»
-WG5	Belden 9841 + battery comms cable	«cores»	CAN bus	-GB1 BMS	51	«m»
-WG6	USB cable	USB	-KE3	-TF1	11, 51	«m»
-WG7	Ethernet patch cable	RJ45	-KE3 enp2s0	-KE4 IN	11	«m»
-WG8	Ethernet patch cable	RJ45	-KE4 OUT	-BG4	11	«m»
-WG9	Ethernet cable (scanner data)	M12 / RJ45	-KE3 enp1s0	-BG1	11, 41	«m»
-WG10	Pendant cable	«5 cores»	-KE4:DI03–DI06, 1N24	+PD	60	«m»

Notes

1. Single-core panel wiring is not listed: AWG22 blue (control, safety) and NYAF black (48 V) inside +EB and to the front panel +FR. It is identified by the potential or signal name on each wire marker (page 3).
2. Lengths are to be measured on the built vehicle.

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0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

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Parts list: all devices shown in this circuit diagram

Tag	Description	Maker	Part number	Rating	Location	Pages	Safety
-GB1	Battery with battery management system (BMS on CAN)	«MAKER»	«PART»	48 V, «Ah»	+BT	20, 51	
-QA1	Circuit breaker, supply disconnecting device (main switch)	«MAKER»	«PART»	2P 32 A DC	+EB	20	
-FC1	Circuit breaker, feeder left motor driver	«MAKER»	«PART»	1P 10 A DC	+EB	20	
-FC2	Circuit breaker, feeder right motor driver	«MAKER»	«PART»	1P 10 A DC	+EB	20	
-FC3	Circuit breaker, 24 V bus	«MAKER»	«PART»	2P 10 A DC	+EB	30	
-FC4	Circuit breaker, 12 V bus	«MAKER»	«PART»	2P 10 A DC	+EB	32	
-TA1	Motor driver, left (CANopen, HWTO)	Oriental Motor	BLVD-KRD	48 V DC, 400 W	+EB	21, 42, 50, 51	Y
-TA2	Motor driver, right (CANopen, HWTO)	Oriental Motor	BLVD-KRD	48 V DC, 400 W	+EB	21, 42, 50, 51	Y
-TA3	DC/DC converter 48 → 24 V, isolated	Mean Well	SD-150C-24	24 V 6.25 A (150 W)	+EB	20, 30	
-TA4	DC/DC converter 48 → 12 V, isolated	Mean Well	«PART»	12 V 8.3 A (100 W)	+EB	20, 32	
-MA1	Motor with gearhead 1:30 and holding brake, left	Oriental Motor	BLMR6400SKM-GFV-B	400 W	+VH	21	
-MA2	Motor with gearhead 1:30 and holding brake, right	Oriental Motor	BLMR6400SKM-GFV-B	400 W	+VH	21	
-KE1	Safety controller main module	SICK	FX3-CPU0	24 V DC	+EB	43	Y
-KE2	Safety controller I/O module, 8 in / 4 out	SICK	FX3-XTIO84002	24 V DC	+EB	40–44	Y
-KE3	Vehicle controller, industrial PC	«MAKER»	«PART»	12 V DC	+EB	11, 32, 51	
-KE4	I/O module, 16 DI / 12 relay DO, Ethernet	«MAKER»	CK-EC5163	24 V DC	+EB	30, 60–62	
-TF1	USB-to-CAN interface	«MAKER»	«PART»	USB	+EB	11, 51	
-SJ1	Emergency stop push-button, 2 NC, positive opening, turn to release	«MAKER»	«PART»		+FR	40	Y
-SJ2	Push-button START, 1 NO	«MAKER»	«PART»		+FR	60	
-SJ3	Push-button RESET, blue, 2 NO	«MAKER»	«PART»		+FR	40, 60	Y
-SJ4	Selector switch MANUAL / AUTO, 2 positions, 2 NO	«MAKER»	«PART»		+FR	40, 60	Y
-SJ5	Switch MANUAL ARM	«MAKER»	«PART»		+FR	60	
-SJ6	Pendant button FORWARD, 1 NO	«MAKER»	«PART»		+PD	60	
-SJ7	Pendant button REVERSE, 1 NO	«MAKER»	«PART»		+PD	60	
-SJ8	Pendant button LEFT, 1 NO	«MAKER»	«PART»		+PD	60	
-SJ9	Pendant button RIGHT, 1 NO	«MAKER»	«PART»		+PD	60	
-BG1	Safety laser scanner, front	SICK	nanoScan3 «type»	24 V DC	+VH	30, 41	Y
-BG3	Magnetic line sensor (CAN)	SICK	MLSE «type»	24 V DC	+VH	31, 51	
-BG4	RFID reader (Ethernet)	«MAKER»	«PART»	12 V DC	+VH	11, 32	
-PJ1	Audio horn, 2 sounds	PATLITE	«PART»	24 V DC	+VH	30, 61	
-PH1	Display	«MAKER»	«PART»	24 V DC	+FR	31	
-GQ1	Fan	NMB-MAT	«PART»	24 V DC	+EB	31	
-GQ2	Fan	NMB-MAT	«PART»	24 V DC	+EB	31	
-GQ3	Fan	NMB-MAT	«PART»	24 V DC	+EB	31	
-XE1	Chassis earth point (PE)				+EB	21	

Notes

Safety = Y: the part carries a safety function (see page 44, SF1 … SF5). Values in red are placeholders or uncertain; fill them before issue.

DRAFT — for TÜV SÜD preliminary review — not for production

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Legal owner

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